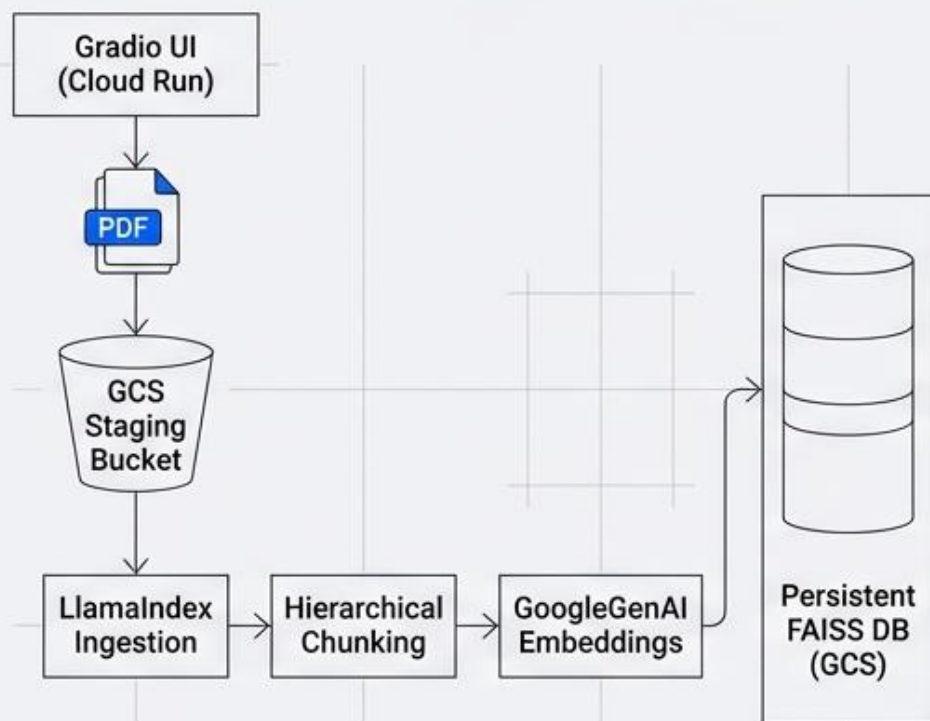
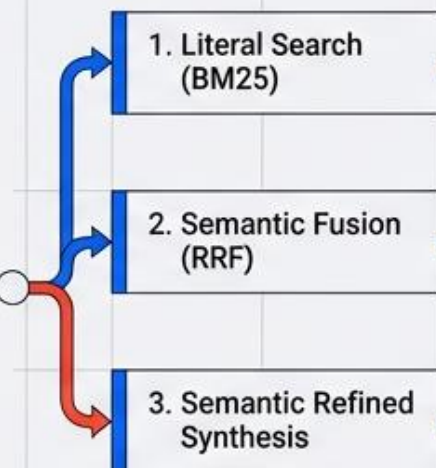


Distributed RAG Architecture & Multi-Agent Research Orchestrator

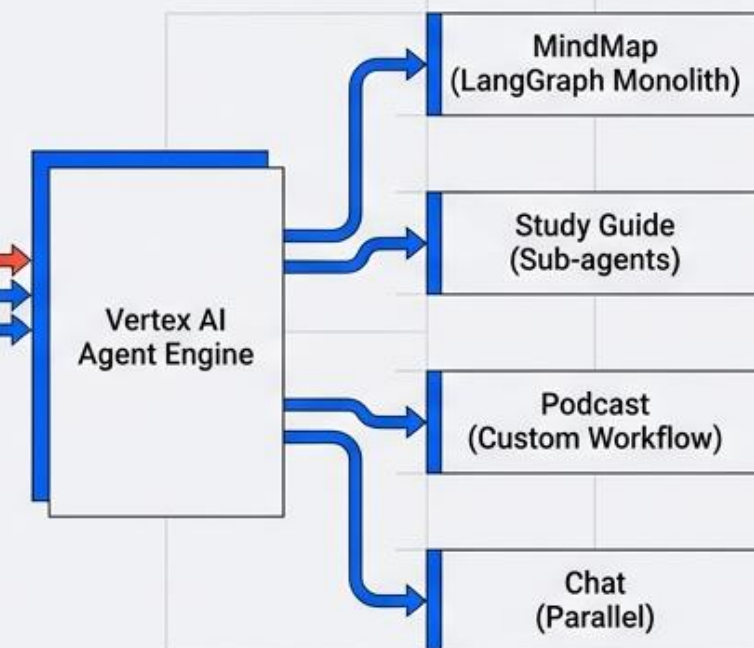
Ingestion & State



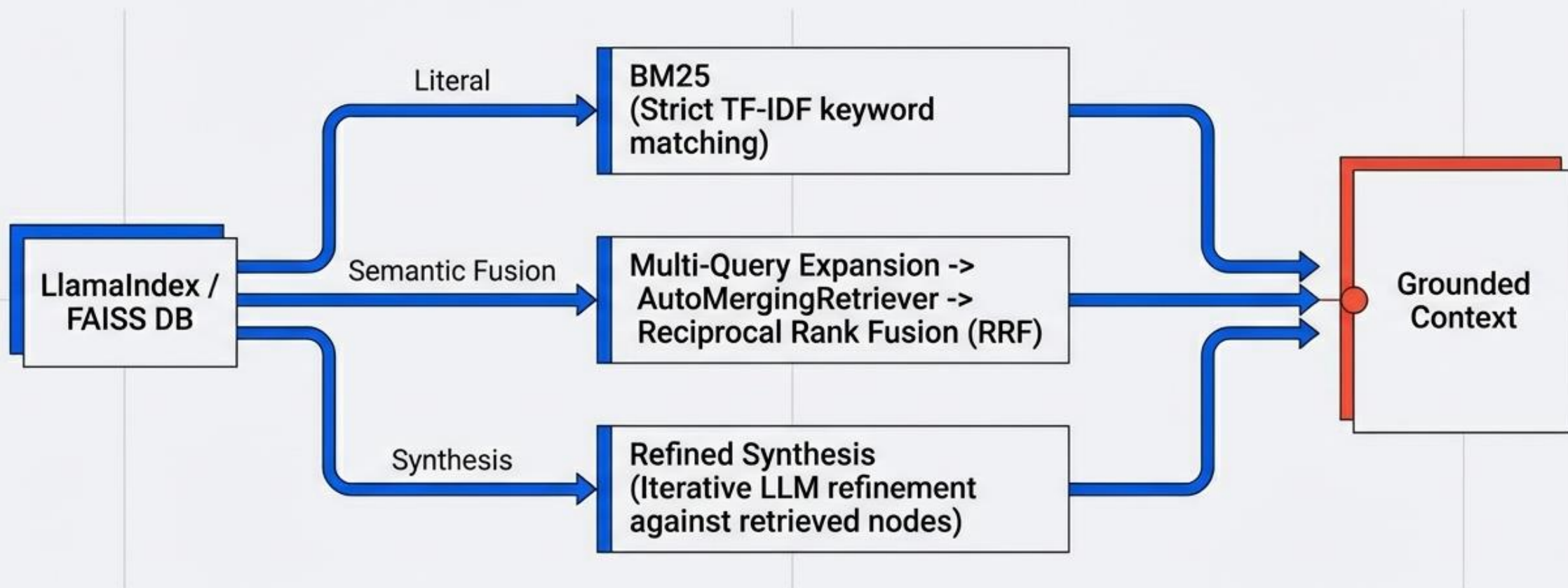
Retrieval (The 3 Paths)



Intelligence (Vertex AI)



Grounded Retrieval: Eliminating Hallucinations via Tri-Path Search



Tech Stack

LlamaIndex

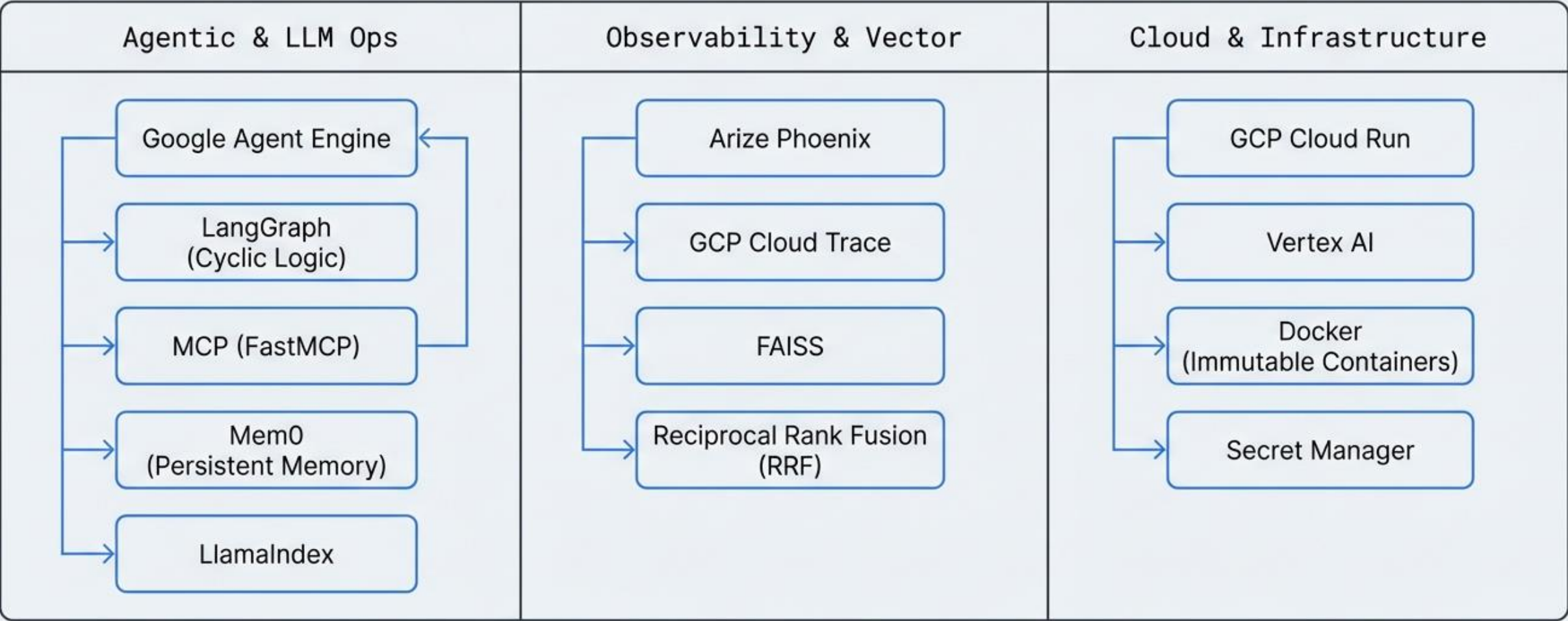
GoogleGenAI Embeddings

FAISS

BM25

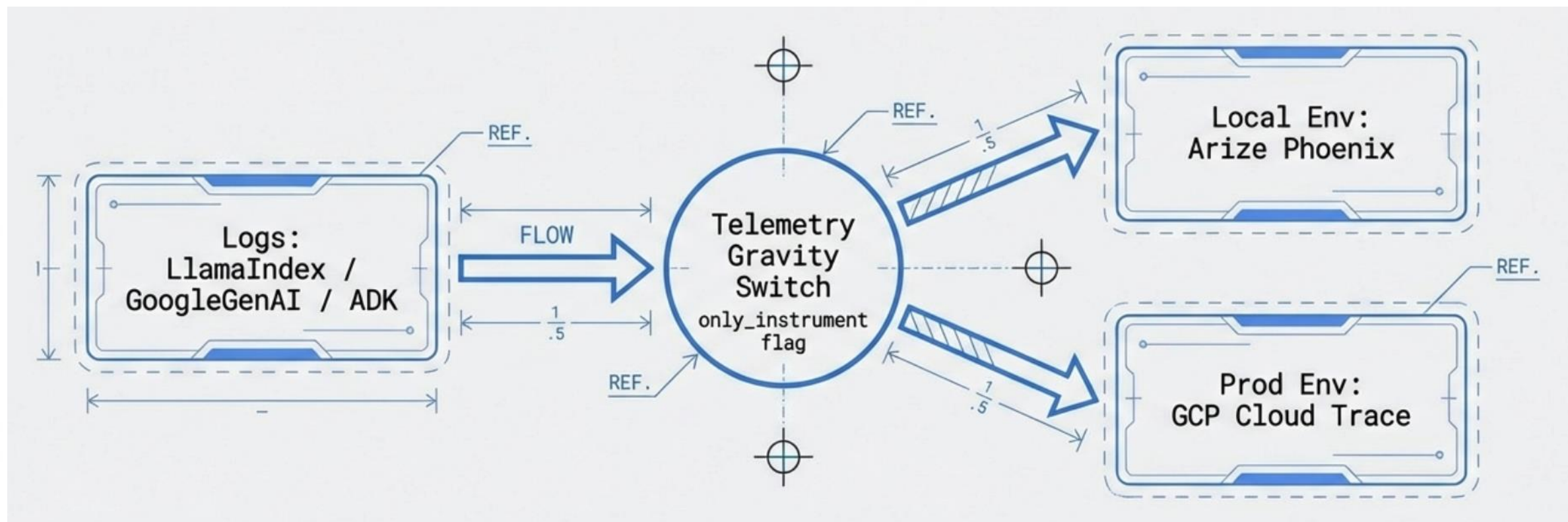
RRF

The Technology Stack: Integration & Polyglot Frameworks



Unifying LangGraph cyclic logic with Google's ADK. Infrastructure relies on immutable Docker containerization via Artifact Registry and GCP serverless ecosystems.

Telemetry Gravity: Unified Observability Pipeline



Abstracted instrumentation using an 'only_instrument' flag. The exact same code pipes granular spans to different destinations based on the environment.

Quality Assurance: High-Fidelity Custom Evaluations



Standard ADK Registry Limitation.
(Built-in evaluator crashed on
Custom Workflow Agents).



Custom JSON Rubric Evaluator.

Engineered an “LLM-as-a-Judge” pipeline against manual JSON rubrics. **Replaced 287 lines of broken framework code with a 10-line Python iterator to validate tool execution sequences.**

FinOps Governance: The 'Nuclear Teardown' Protocol



The Challenge

Vertex AI Agent Engine bills continuously for "warm" idle instances, unlike Cloud Run which scales to zero.

The Protocol

- ✓ 1. Manually scale Cloud Run to zero.
- ✓ 2. Delete Vertex AI reasoning engines via SDK.
- ✓ 3. Purge Artifact Registry builds.
- ✓ 4. Revoke IAM permissions.

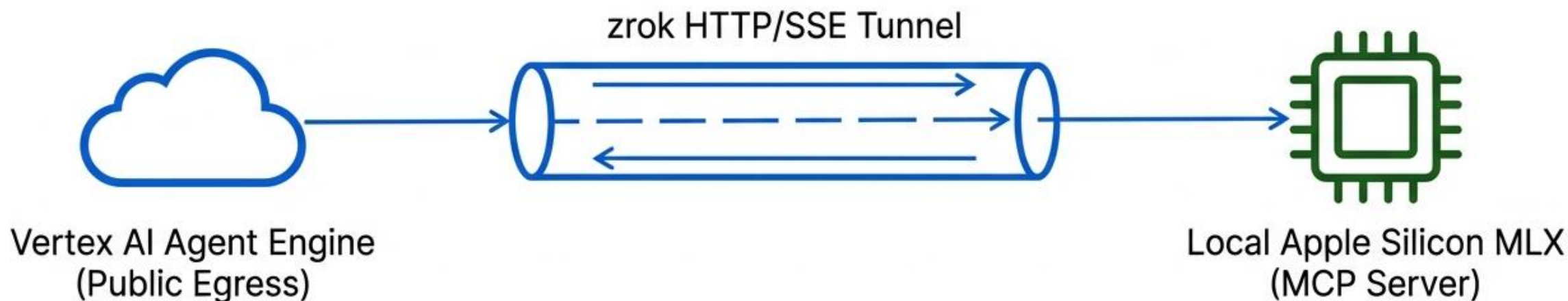
Tech Stack

GCP Billing

Vertex AI SDK

Cloud Run

Hybrid Networking: The Cloud-to-Edge Bridge



- Context: Vertex AI Linux VMs cannot compile Apple instructions required for the Podcast agent's audio synthesis.
- Solution: Migrated local MCP server to use Server-Sent Events (SSE) and exposed via a stable zrok tunnel, routing cloud reasoning to isolated local hardware.

Security, Deployment, and Storage Architecture

Structural Security: Zero-Trust Architecture



Zero-Trust

IAM Least Privilege

Frontend UI restricted to 'run.invoker';
Backend restricted to 'storage.objectViewer'

Secret Vaulting

API keys bound to Compute Service
Accounts via 'secretAccessor'

Signed URL Delivery

TTS server generates Cryptographic v4
Signed URLs for secure ephemeral audio.

Deployment Physics: Payload & Compilation



8MB Payload Diet

Aggressive ".ae_ignore"
shrinking deployments by 99%



Cross-Arch Compilation

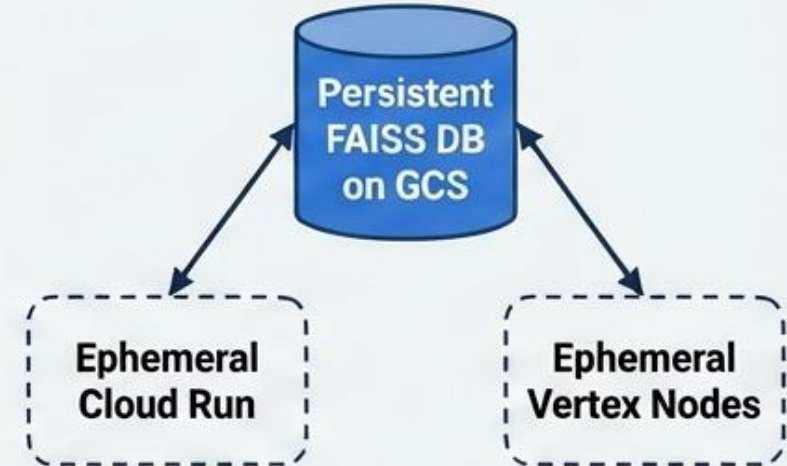
Enforced "--platform
linux/amd64" Docker digests



Serverless Health-Checks

Tuned TCP startup probes and
"0.0.0.0" port bindings

Solved: Stateful Storage in Ephemeral Serverless AI



Decoupled storage from compute,
synchronizing FAISS indexes directly
against a shared Google Cloud Storage
staging bucket.

Tech Stack

GCP Secret Manager

IAM

Google Cloud Storage (GCS)

Docker

.ae_ignore

Cloud Run Probes

Artifact Registry

FAISS

Cloud Run